
REPORT
ON
DESIGN & DEVELOPMENT OF IOT BASED HEALTH
MONITORING SYSTEM
EEN-300: INDUSTRY ORIENTED PROBLEM
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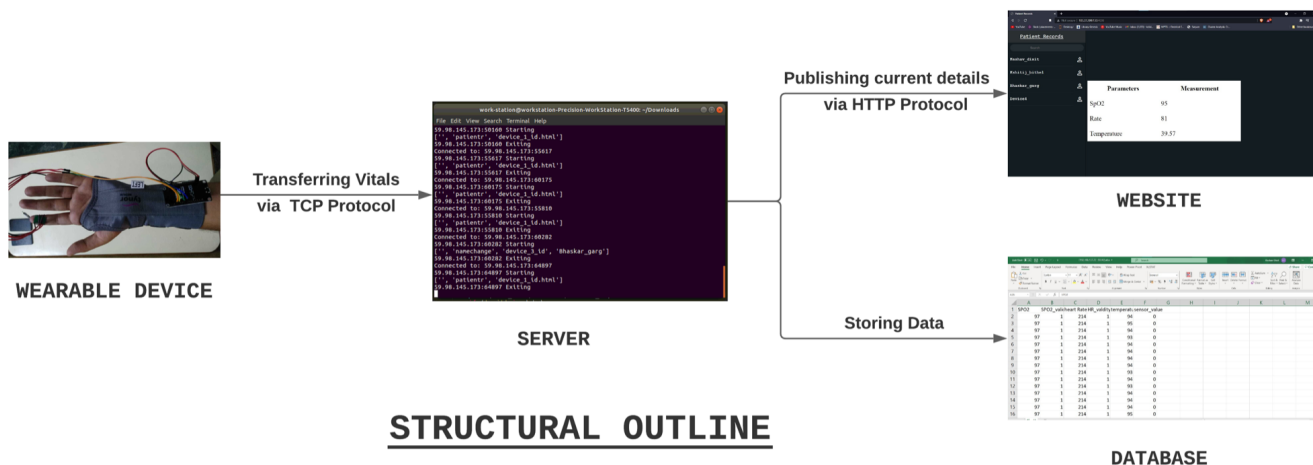
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I. ABSTRACT

The widespread pandemic has forced us to ponder about developing new and better ways for health monitoring for safer and easier diagnosis of patients.

With the guidance of associated doctor Dr Alok Anand, we found that 4 essential parameters extracted from the body are used to diagnose a large group of diseases, namely Respiration Rate (RR), SPO2, Pulse Rate (PR), and Body Temperature (BT). For ergonomic monitoring of these parameters we aimed to create a small wearable device that can monitor these parameters in real-time and can communicate them distantly in a doctor-friendly manner.

As an extension, we aim to analyse the effect of YAM(Yoga and Meditation) on patients, as research performed on YAM suggests that it helps in relieving stress, supporting good health habits, and improving mental/emotional health. It also supports people with some chronic diseases like COPD or Asthma to manage their condition in addition to treatment programs. We also aim to analyse the effect on YAM on COVID-19 patients and also on post-COVID complications.



II. INTRODUCTION

As per WHO, the most common symptoms of COVID-19 are fever, dry cough, and tiredness. The primary scanning for the suspect of COVID-19 is done on the basis of the body temperature of a person. COVID-19 is a respiratory disease; therefore, shortness of breath is also a symptom of the disease. The blood oxygen level is an indicator of lung functioning and provides an early warning for the kinds of breathing problems associated with COVID-19 pneumonia.

Here we aim to design and develop a monitoring system, which could continuously record the vital clinical parameters (respiration rate, blood oxygen level, heart rate, and body temperature), giving the diagnostic aid for finding the state of COVID-19 infection. The system has two parts; the first one is an electronic mobile unit which is in contact with the human body with the ability to communicate through a wireless network; the second one is a listener (server) that is used to receive and store the data.

The data is acquired by the server and updated continuously on a website for real-time remote monitoring by a doctor. It would also be beneficial to analyze and study the effect of YAM on these vital health parameters. So the server logs the data. These recorded parameters can also be used to categorize the patients into normal, alert, or severe categories, and in the monitoring the movement of quarantined persons using the mobile unit.



Minimum Viable Product

III. MOTIVATION

As discussed in Introduction the ongoing pandemic created an atmosphere of fear and anguish in the general population, and especially among the doctors. In COVID it is very necessary to maintain physical distance with the patient. But being a doctor it is really difficult to monitor the patient with distance.

The Department of Science and Technology(DST) also identified this issue and took the initiative to develop necessary hardware for this along with Dr Vishal Kumar. We, in the team, worked to give this funded project a face of final usable product.



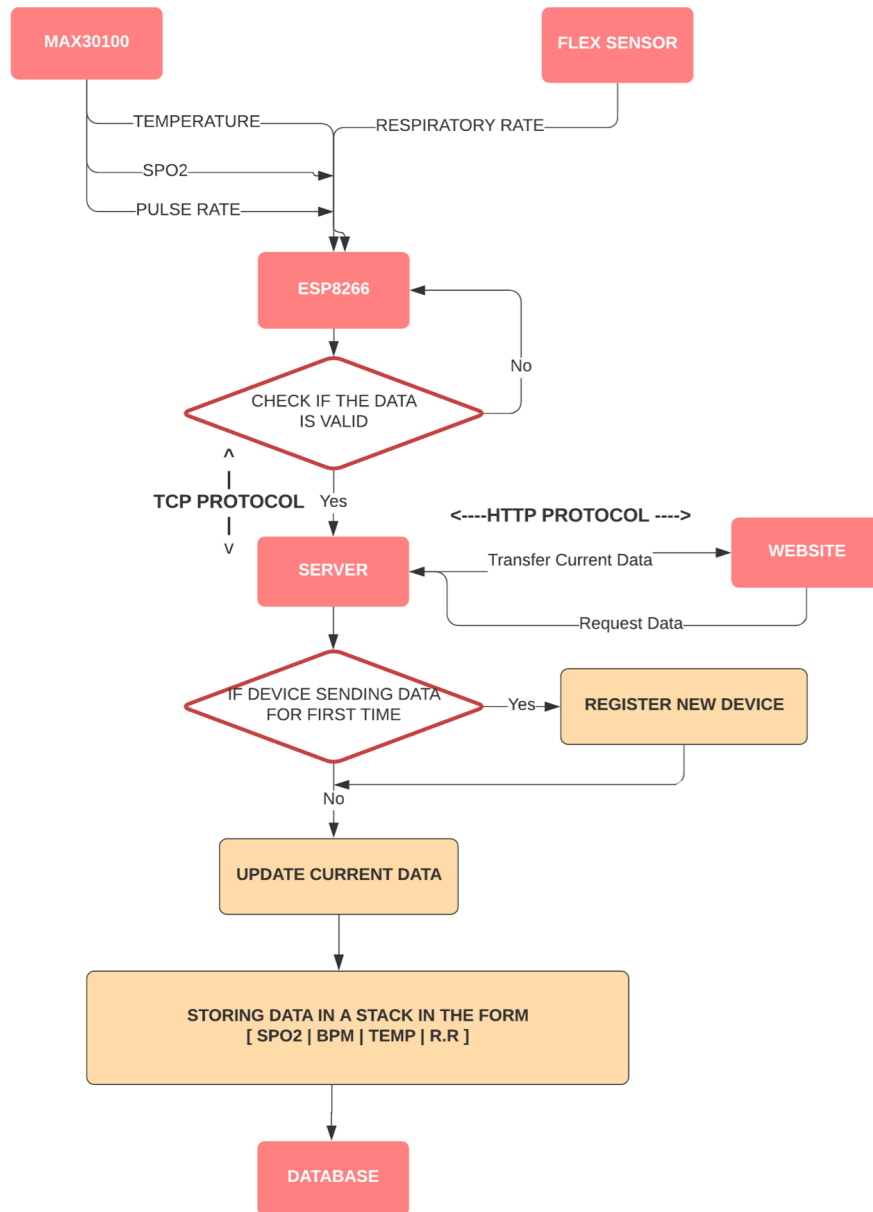
Current Monitoring Station which requires presence of both doctor and patient in the same room.

IV. OBJECTIVES

- Design and development of electronic wearable health device (WHD) to monitor vital clinical parameters useful for COVID-19 diagnosis
- WHD with the ability of wireless communication
- Development of a Unix/Linux based listener for receiving and storing the data for remote monitoring of the parameters
- Analysis of the data and reporting inference about the influence of YAM on COVID-19 related health parameters

V. TECHNICAL DETAILS

The system has three units, a Wearable Health Device (WHD), a receiver to collect the data related to the health parameters AND the Website.



Overall Dataflow

1. Wearable Health Device (WHD)

The Device at the Patient's end which interfaces the sensors and measures the selected health parameters(RR, SPO2, BT, PR). R The device is divided into 2 modular parts : Wrist band for SPO2, BT, PR and chest belt for RR measurement. These are further described below

1.1. Implementation of Wrist band

Calculation of SPO2, Pulse Rate, Body Temperature is performed using MAX30100 sensor which uses infrared absorption technique for measurements. It communicates with the processor through I2C communication. An ESP8266 development board is used as the central microcontroller, it receives the data from all the sensors, does the necessary filtering and uses an on board wifi module to send the data to a remote server over the internet. The wristband houses the nodeMCU , Max30100, and battery. The respiratory belt can be attached to the wrist band via a connector, leading to a modular design.

1.2. Implementation of Chest belt for Respiratory monitoring

Calculation of respiration rate is performed using a strain gauge, also known as flex sensor. The principle of flex sensor is based on changing resistance on bending the strip. When the flex sensor is bent the resistance changes and the change is monitored using a microcontroller.

A flex sensor is attached on an elastic based chest belt. While respiration the chest expands and contracts. The expansion and contraction is reflected in the strain gauge attached to the chest. And hence the respiration rate can be calculated.

In this way, a modular chest belt is prepared which can be connected to the wrist band and we can get the respiration rate whenever required by just attaching the chest belt to the band.

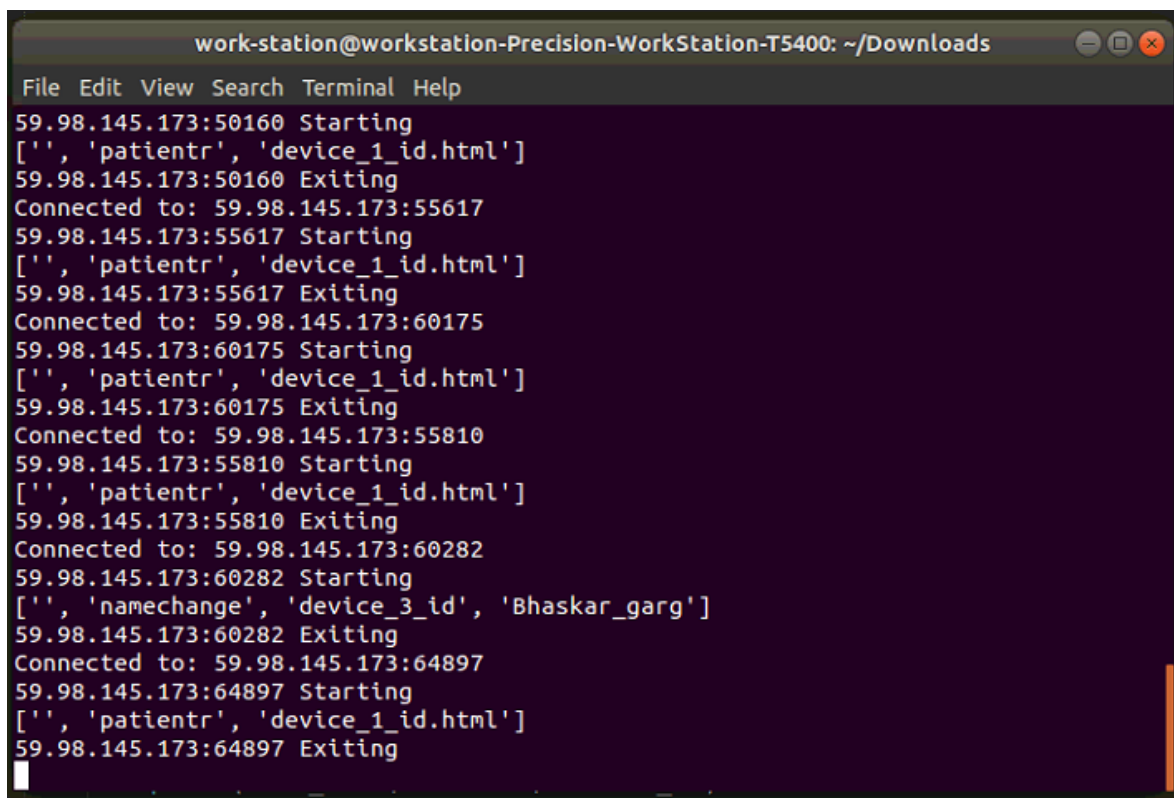


Wearable Hardware Device

2. Server

The server is developed using the Python socket module. It is hosted over a static IP address and listens to requests made on the same. A WHD client distinguishes itself by sending a unique keyword which when received, a function is called. The function takes the data sent by the client and stores it in a list, updates the latest value of parameters on the website, and makes an Excel file of the data received. When someone tries to access the website, a “HTTP GET” request is generated and the server responds to it by sending the updated HTML script. Different Request which are served through the server are:-

- Request for Website
- Request for the data of a particular Patient.
- Request for the Replacement of the Device ID with the Patient name.



```
work-station@workstation-Precision-WorkStation-T5400: ~/Downloads
File Edit View Search Terminal Help
59.98.145.173:50160 Starting
['', 'patientr', 'device_1_id.html']
59.98.145.173:50160 Exiting
Connected to: 59.98.145.173:55617
59.98.145.173:55617 Starting
['', 'patientr', 'device_1_id.html']
59.98.145.173:55617 Exiting
Connected to: 59.98.145.173:60175
59.98.145.173:60175 Starting
['', 'patientr', 'device_1_id.html']
59.98.145.173:60175 Exiting
Connected to: 59.98.145.173:55810
59.98.145.173:55810 Starting
['', 'patientr', 'device_1_id.html']
59.98.145.173:55810 Exiting
Connected to: 59.98.145.173:60282
59.98.145.173:60282 Starting
['', 'namechange', 'device_3_id', 'Bhaskar_garg']
59.98.145.173:60282 Exiting
Connected to: 59.98.145.173:64897
59.98.145.173:64897 Starting
['', 'patientr', 'device_1_id.html']
59.98.145.173:64897 Exiting
```

Server in Action

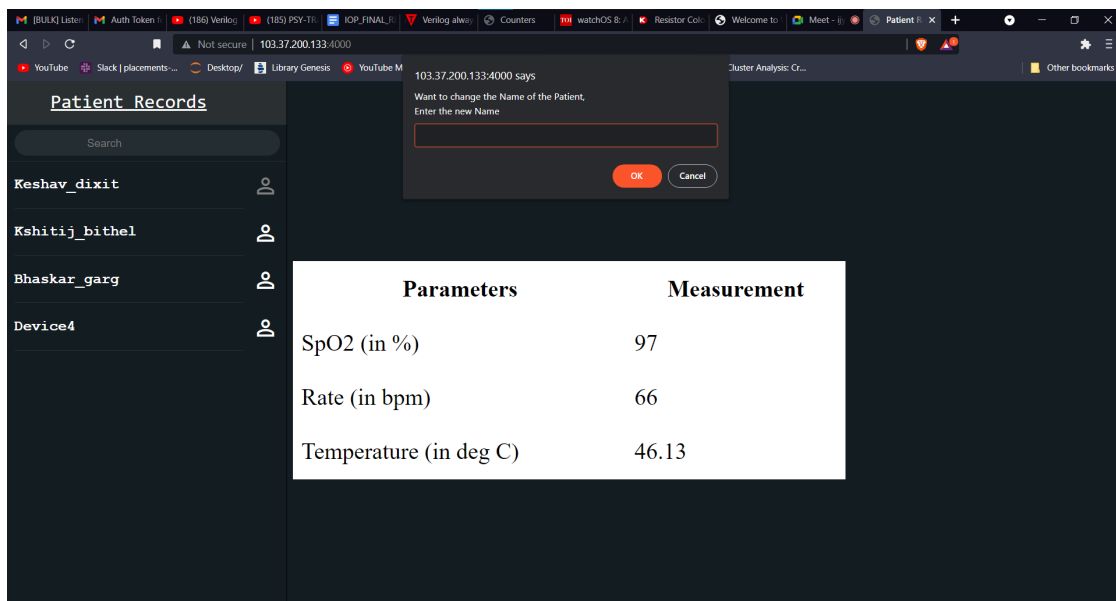
We have designed the server using Multithreading such that it can handle multiple requests at the same time and there can be multiple clients connected to it. Whenever a new client is connected, the programme creates a separate thread to serve the client resulting in each client running on a new Thread parallelly with other threads and the Main Programme thread. Similarly when any of the Requests explained by the browser are made then it is served in a separate thread without disturbing the previous running the tasks.

3. Website

This is the part which interacts with the User. This site is responsible for conveying all the data of the patient to the Doctor and should also allow the Doctor or Nurse to make necessary changes to the site.

Hence the Website serves these Purposes. It displays the data in Whatsapp type template where the Patient names can be clicked on to open the information related to that Patient. Further there is a “contact icon” alongside the name which can be clicked upon to open a dropdown space to provide the name. Further there is also a Search Bar provided so that anyone can search the name from the list.

The Patient Data refreshes in every 10 seconds if the WHD is connected, else if not connected the last received data is continuously displayed.



Website in Action

Its main features are:-

- Has a search bar to search Patients name
- Allows doctors to change names corresponding to devices.
- Refreshes the data of the Patient every 10 seconds.
- Easy to use Whatsapp like interface.

VI. RESULTS

We have made a hardware prototype capable of transmitting all the required parameters(RR, PR, BT, SPO2) to the static server based on python sockets. After testing our prototype, we found that the Max30100 sensor readings were inaccurate near the chest so we had to build a glove with a finger mount(for SPO2,BT,PR) and an accompanying chest belt(for RR).

Our Server is able to host multiple clients and also caters to multiple features available at the user's end. Such as renaming the Patient's name for using the device on multiple Patients. Also the server logs the data being received from the device so that the data can be used for further analysis.

Our Website is Fully functional and Provides the basic features for the effective usability of the device. It provides easy navigation through whatsapp like interface, allows us to search a patient, and also deliver us the facility of changing the name of the patient through a very simple Interface.

VII. DISCUSSION

There are several Wearable Health Monitoring Devices available in the market and the applications vary from monitoring of Temperature, Blood Oxygen Content, Pulse Rate, ECG etc. Under the supervision of Dr Alok Anand we could identify a gap in available health products. Respiration Rate being the most important parameter for heart and lung related monitoring, there is no WHD available that can measure the respiration rate.

Respiration rate being a parameter of uttermost importance according to medical professionals. We included precise calculation of respiration rate in the form of a wearable chest belt.

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